INCH - POUNDS

MS27395C
12 December 2011
SUPERSEDING
MS27395B
24 September 2003

DETAIL SPECIFICATION SHEET

ELBOW, FLARELESS, TUBE TO HOSE - 45° SWIVEL NUT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-27272.

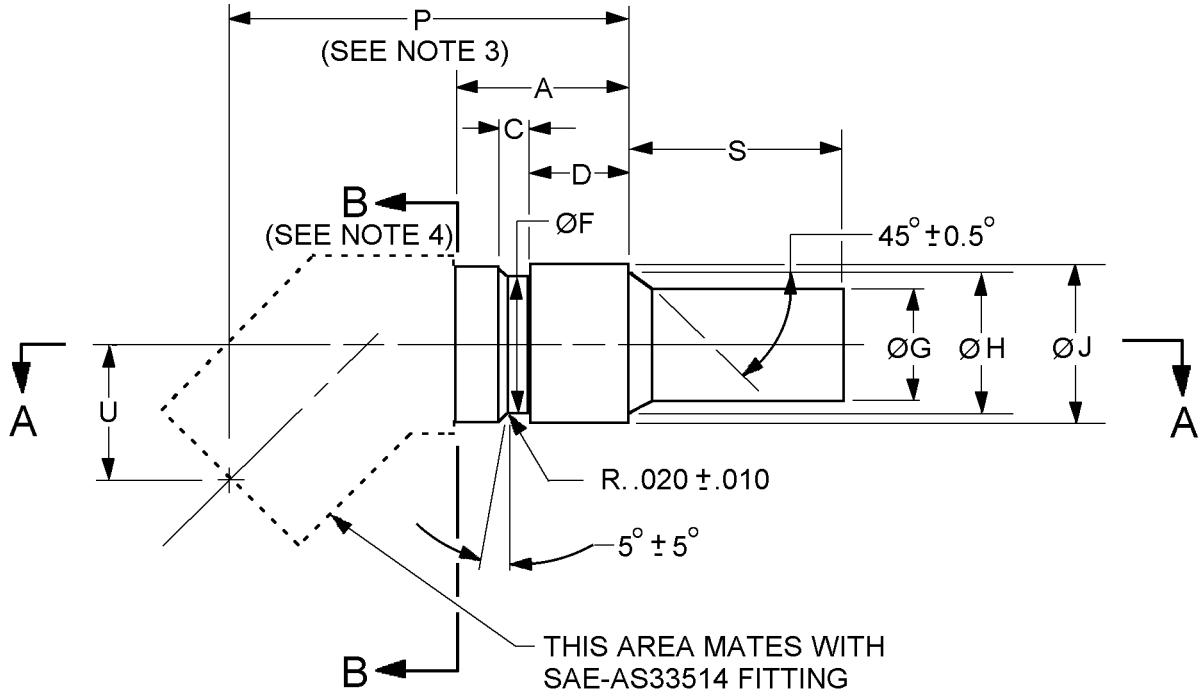
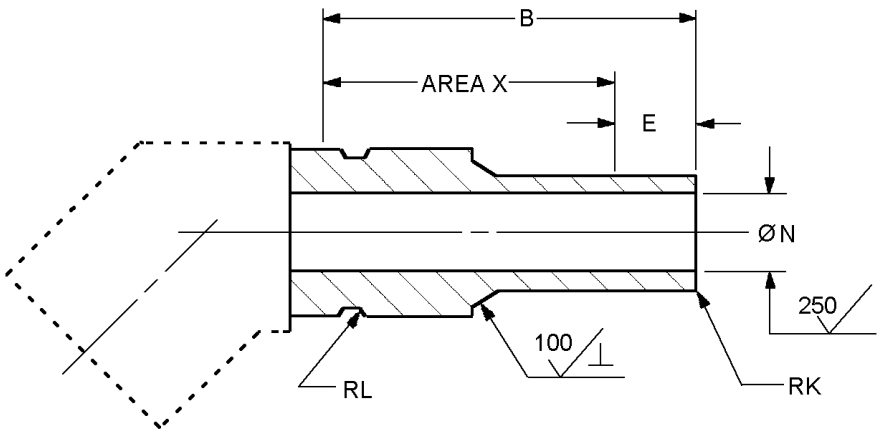
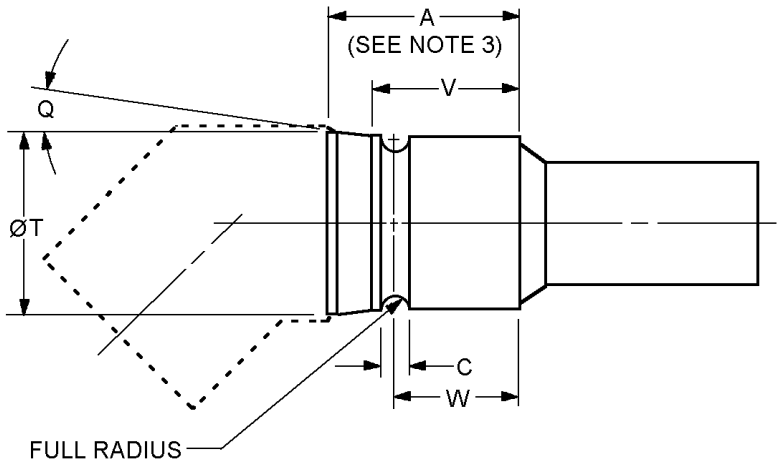


FIGURE 1. Elbow dimensions and configuration.

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SECTION A - A



-10 THROUGH -24 SIZE ONLY

FIGURE 1. Elbow dimensions and configuration - Continued.

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Dash number		A min (see note 3) Inches (mm)	B min Inches (mm)	C Inches (mm)		D +.005 -.000 Inches (mm)	E Inches (mm)	
Steel	Alum							
-3/-4C	---	.515 (13.08)	.870 (22.10)	.106 (2.69)	±.010 (0.25)	.307 (7.80)	.25 (6.35)	±.05 (1.3)
-4C	---							
-5C	---	.520 (13.21)	.890 (22.61)	.116 (2.95)		.322 (8.18)		
-6C	---	.555 (14.10)						
-8C	-8D	.630 (16.00)	1.030 (26.16)	.140 (3.56)	±.020 (0.51)	.335 (8.51)	.32 (8.1)	±.12 (3.0)
-10C	-10D	.654 (16.61)	1.130 (28.70)	.098 (2.49)	+.004 (0.10) -.000	---	.35 (8.9)	±.15 (3.80)
-12C	-12D	.755 (19.18)	1.240 (31.50)	.128 (3.25)	+.005 (0.13) -.000	---		
-16C	-16D	.831 (21.11)	1.340 (34.04)			---	.39 (9.9)	±.19 (4.8)
-20C	-20D	.881 (22.38)	1.570 (39.88)			---	.48 (12.2)	±.28 (7.1)
-24C	-24D	1.035 (26.29)	1.720 (43.69)			---	.50 (12.7)	±.30 (7.6)

FIGURE 1. Elbow dimensions and configuration - Continued.

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Dash number		F dia. Inches (mm)		G dia. +.005 (0.13) -.000 Inches (mm)	H dia. ±.005 0.13) Inches (mm)	J dia. +.005 (0.13) -.000 Inches (mm)
Steel	Alum					
-3/-4C	---	.288 (7.32)	+.005 (0.13) -.000	.205 (5.21)	.295 (7.49)	.373 (9.47)
-4C	---					
-5C	---	.316 (8.03)		.268 (6.81)	.360 (9.14)	.416 (10.57)
-6C	---	.391 (9.93)		.330 (8.38)	.425 (10.80)	.491 (12.47)
-8C	-8D	.497 (12.62)		.426 (10.82)	.530 (13.46)	.616 (15.65)
-10C	-10D	.586 (14.88)		.526 (13.36)	.625 (15.88)	.706 (17.93)
-12C	-12D	.674 (17.12)		.650 (16.61)	.760 (19.30)	.826 (20.98)
-16C	-16D	1.001 (25.43)	+.008 (0.20) -.000	.900 (22.86)	1.040 (26.41)	1.150 (29.21)
-20C	-20D	1.255 (31.88)	+.005 (0.13) -.000	1.151 (29.24)	1.275 (32.39)	1.405 (35.69)
-24C	-24D	1.490 (37.85)		1.401 (35.59)	1.550 (39.37)	1.635 (41.53)

FIGURE 1. Elbow dimensions and configuration - Continued.

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Dash number		K Inches (mm)		L ±.004 (0.10) Inches (mm)	M (see note 3) min Inches (mm)	N dia. Inches (mm)		
Steel	Alum					Steel	Alum	
-3/-4C	---	.015 (0.38)	+.005 (0.13) -.000	.036 (0.91)	.485 (12.32)	.161 (4.09)	---	+.005 (0.13) -.000
-4C	---						---	
-5C	---			.040 (1.02)	.501 (12.73)	.224 (5.69)	---	
-6C	---				.521 (13.23)	.261 (6.63)	---	
-8C	-8D	.020 (0.51)		.047 (1.19)	.583 (14.81)	.345 (8.76)	.345 (8.76)	+.006 (0.15) -.000
-10C	-10D		±.005 (0.13)	---	.620 (15.75)	.440 (11.18)	.440 (11.18)	
-12C	-12D	.030 (0.76)		---	.720 (18.29)	.560 (14.22)	.560 (14.22)	
-16C	-16D			---	.796 (20.22)	.828 (21.03)	.828 (21.03)	
-20C	-20D			---	.846 (21.49)	1.058 (26.87)	1.058 (26.87)	
-24C	-24D	.035 (0.89)		---	1.000 (25.40)	1.253 (31.83)	1.282 (32.56)	+.005 (0.13) -.000

FIGURE 1. Elbow dimensions and configuration - Continued

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Dash number		N Inches (mm)			P Inches (mm)		Q max	S Inches (mm)	
Steel	Alum	Steel	Alum						
-3/-4C	---	.161 (4.09)	---	+.005 (0.13) -.000	1.227 (31.17)	±.020 (0.51)	---	.480 (12.19)	±.010 (0.25)
-4C	---		---		1.219 (30.96)		---		
-5C	---	.224 (5.69)	---		1.258 (31.95)		---		
-6C	---	.261 (6.63)	---		1.361 (34.57)		---		
-8C	-8D	.345 (8.76)	.345 (8.76)	+.006 (0.15) -.000	1.877 (47.68)		---	.600 (15.24)	
-10C	-10D	.440 (11.18)	.440 (11.18)		1.674 (42.52)	±.035 (0.89)	---	.650 (16.51)	±.015 (0.38)
-12C	-12D	.560 (14.22)	.560 (14.22)		2.114 (53.70)		15.5º	.675 (17.15)	±.025 (0.64)
-16C	-16D	.828 (21.03)	.828 (21.03)		2.188 (55.58)		10.5°	.730 (18.54)	
-20C	-20D	1.058 (26.87)	1.058 (26.87)		2.445 (62.10)		15.5º	.935 (23.75)	
-24C	-24D	1.253 (31.83)	1.282 (32.56)	+.005 (0.13) -.000	2.849 (72.36)			.980 (24.89)	

FIGURE 1. Elbow dimensions and configuration - Continued.

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Dash number		T dia. max Inches (mm)	U Inches (mm)		V Inches (mm)		W +.005 (0.13) -.000 Inches (mm)	
Steel	Alum						Steel	Alum
-3/-4C	---	---	.432 (10.97)	±.020 (0.51)	---	---	---	---
-4C	---	---	.424 (10.77)		---	---	---	---
-5C	---	---	.448 (11.38)		---	---	---	---
-6C	---	---	.526 (13.36)		---	---	---	---
-8C	-8D	---	.622 (15.80)		---	---	---	---
-10C	-10D	---	.725 (18.42)	±.035 (8.98)	---	±.010 (0.25)	.420 (10.67)	.427 (10.85)
-12C	-12D	.900 (22.86)	.800 (20.32)		.625 (15.88)		.500 (12.70)	.500 (12.70)
-16C	-16D	1.190 (30.23)	.854 (21.69)		.670 (17.02)		.545 (13.84)	.545 (13.84)
-20C	-20D	1.485 (37.72)	.962 (24.43)		.695 (17.65)		.565 (14.35)	.571 (14.50)
-24C	-24D	1.750 (44.45)	1.155 (29.34)		.795 (20.19)		.665 (16.89)	.665 (16.89)

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Use A dimension when the adjacent diameter to the left of plane B - B is greater than J dimension. When the adjacent diameter is equal to or less than J dimension, M dimension may be used in place of A dimension.
4. Any design of the elbow to the left of plane B - B is acceptable provided the dimensions P and U and the requirements of this specification sheet and the procurement specification are met.
5. Any forged design is acceptable for sizes -3/-4 through -8.
6. Any bent tube design is acceptable for sizes -10 through -24.
7. The inside diameter (ID) of the elbow, sizes -10 through -24, shall not be less than the ID of the nipple end of the elbow.
8. Ovality shall not exceed 7.5 percent of nominal tubing outside diameter (OD).
9. Unless otherwise specified, break or radius all corners +.005, -.000 inch (+0.13 -0.00 mm).
10. All diameters within length A plus S must be concentric within .005 inch (0.13 mm) full indicator movement.
11. Remove all burrs and slivers.
12. Surface roughness unless otherwise specified, maximum surface roughness shall not exceed 125μ inch. R_a in accordance with ASME B46.1.

FIGURE 1. Elbow dimensions and configuration - Continued.

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REQUIREMENTS:

Intended use. This part is a component of MS27389 for sizes -3/-4C through -8, and MS27387 for sizes -10 through -24. This is a design standard for manufacturing purposes. The item is only procured as an integral part of adapter assemblies.

Dimensions and configuration, see figure 1.

Material:

Corrosion-resistant steel, class 304 or 321 cold drawn or cold rolled, in accordance with SAE-AMS-5639 or SAE-AMS-5645.

Aluminum alloy, 6061-T6 or T651, in accordance with SAE-AMS-4117 or alloy 7075-T73, 7075-T7351 in accordance with SAE-AMS-QQ-A-225/9 or alloy 7055-T7351 in accordance with SAE-AMS-4142.

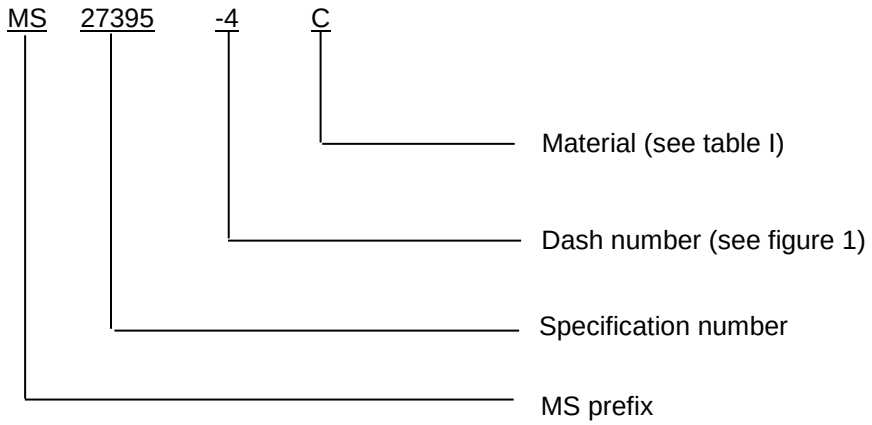
Finish:

Corrosion-resistant steel, passivate in accordance with SAE-AMS-2700. Dry-film lubricate area X, see figure 1, with lubricant conforming to SAE-AS1701. No overspray allowed, dry-film same area for -10 thru -24.

Aluminum alloy, anodize in accordance with MIL-A-8625, type II, dye yellow.

Order of precedence. This specification takes precedence over the documents referenced herein. Unless otherwise specified, referenced documents shall be of the issue in effect on the date of solicitation.

Part or Identifying Number (PIN):



PIN example: MS27395-4Cis a 45º elbow assembly that mates with an SAE-AS33514-4 fitting and a .250 inch hose, corrosion resistant steel.

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TABLE I. Material to suffix code.

Suffix code	Material
C	Corrosion-resistant steel
D	Aluminum

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Referenced documents. In addition to MIL-DTL-27272, this document references the following:

MIL-A-8625
MS27387
MS27389
ASME B46.1
SAE-AMS-QQ-A-225/9
SAE-AMS-2700
SAE-AMS-4117
SAE-AMS-4124
SAE-AMS-5639
SAE-AMS-5645
SAE-AS1701
SAE-AS33514

CONCLUDING MATERIAL

Custodians:
Army - AV
Navy - AS
Air Force - 99
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2011-097)

Review activities:
Army - AT, MI
Navy - MC, SA, SH
Air Force - 71

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

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